



GATE SMASHERS

GATE2024

Engineering Mathematics

Topic: Probability and Statistics

Subtopic: Basic Formulas Of Factorial And Permutation

LECTURE - 2

EXPLANATION

Factorial - Product of n natural numbers and denoted by $n!$ or $1n$



eg - $9! = 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$

$9! = 9 \times 8 \times 7 \times 6 \times 5!$

$n! = n(n-1)(n-2)(n-3) \dots 3 \cdot 2 \cdot 1$

Note - $1! = 1$ and $0! = 1$

Example - (i) $\frac{1}{6!} + \frac{1}{7!} + \frac{1}{8!}$

(ii) $6! \times (1 \cdot 3 \cdot 5 \dots 11) \cdot 2^6$

Permutation when all the objects are distinct -

${}^n P_r = \frac{n!}{(n-r)!}$
no. of different objects
object taken at a time.

eg - (i) ${}^{18} P_2$ (ii) ${}^n P_6 = 3 {}^n P_5$ (Find n)

Note - $P(n, n) = {}^n P_n = n!$

ex - There are 3 different rings to be worn in four fingers with atmost one in each finger. In how many ways can this be done?